TANGENT SPACES OF THE STACK OF CURVES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Section artin-section-tangent-spaces (canonical tag 07WY). cited at statement line 4953. Tangent spaces of a functor.
- Section artin-section-inf (canonical tag 07Y6). cited at statement line 4956. Infinitesimal spaces.
- Equation quot-equation-curves-over-proper-spaces (canonical tag 0D50). cited at proof line 4963. The fully faithful embedding.
- Lemma quot-lemma-spaces-tangent-space (canonical tag 0D1K). cited at proof line 4965. Collected card: finiteness for the stack of spaces.

Source reference. Original section 07WY: sources/stacks-chow/tags.txt, source lines 10058–10058. Cited at statement line 4953: definition of tangent spaces.

Source reference. Original section 07Y6: sources/stacks-chow/tags.txt, source lines 10101–10101. Cited at statement line 4956: definition of infinitesimal spaces.

Source reference. Original equation 0D50: sources/stacks-chow/tags.txt, source lines 16470–16470. Cited at statement line 4963: the fully faithful embedding.

Source reference. Original lemma 0D1K: sources/stacks-chow/tags.txt, source lines 16350–16350. Cited at statement line 4965: corresponding finiteness for the stack of spaces.

Lemma 0.2. *Let k be a field and let $x = (X \to \mathrm{Spec}(k))$ be an object of $\mathcal{X} = \mathcal{C}urves$ over $\mathrm{Spec}(k)$.*

(1) *If k is of finite type over $\mathbf{Z}$, then the vector spaces $T\mathcal{F}_{\mathcal{X},k,x}$ and $Inf(\mathcal{F}_{\mathcal{X},k,x})$ (see Artin's Axioms, Section 07WY) are finite dimensional, and*
(2) *in general the vector spaces $T_x(k)$ and $Inf_x(k)$ (see Artin's Axioms, Section 07Y6) are finite dimensional.*

Proof. This is immediate from the fully faithful embedding (0D50) and the corresponding fact for $\mathcal{S}paces'_{fp,flat,proper}$ (Lemma 0D1K). □

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